Simple Interest & Compound Interest

- By Sunil Guptha Yarramalla

↳ **Rate of Interest on Interest:-** Interest to be charged according to a specific term, which is expressed as some percent of the principal /capital/ borrowed money & is called the rate of interest for the fixed period of time.

→ Fixed period of time is either per annum (annually), per half year (half-yearly), per 3 months (quarterly) (or) per month (monthly) .... etc.

↳ **Principle:-** (P) Capital (or) sum of money borrowed for interest.

↳ **Amount:-** (A):- Amount to be returned by adding interest to principle.

↳ **Time period (T) (or) No. of years (n):-** Particular period of time money borrowed.

① **Simple Interest:-** Interest is to be considered on principle only & constant for particular period for 'n' no. of times, Based on question.

$$SI = \frac{P\,T\,R}{100} \Rightarrow P = \frac{100\,SI}{T\,R}$$

$$\Rightarrow R = \frac{100\,SI}{PT} \qquad \boxed{A_1 = P + SI}$$

$$\Rightarrow T = \frac{100\,SI}{PR}$$

A_1 = Amount in SI

Compound Interest!- Interest on interested amount is added consecutively (or) Successively. is **Compound Interest**. [or we can say interest on interest].

$$A_2 = [\text{Amount in CI}] = P\left(1+\frac{R}{100}\right)^n = P + CI$$

$$\Rightarrow CI = P\left(1+\frac{R}{100}\right)^n - P = P\left[\left(1+\frac{R}{100}\right)^n - 1\right]$$

Example table.

Ex:- $P = 1000/-$, $R = 10\%$ P.a,

= Note ④

T (or) n	$SI = \frac{PTR}{100}$	$A_1 = P+SI$	$A_2 = P\left(1+\frac{R}{100}\right)^n$	$CI = A_2 - P$	CI - SI	$A_2 - A_1$
After 1year	$SI = \frac{1000\times1\times10}{100}$ = 100/- (or) = 10% of 1000 = 100/-	= 1000 + 100 = 1100/-	$A_2 = 1000\left(1+\frac{10}{100}\right)^1$ = 1000×1.1 = 1100/-	CI = 1100-1000 = 100/-	100/- 100/- = 0 Note ③	1100 - 1100 = 0
After 2 years	SI = 20% of 1000 = 200 [10%+10%] of P	= 1000 + 200 = 1200/-	$A_2 = 1000\left(1+\frac{10}{100}\right)^2$ $= (1.1)^2 1000$ = 1210/-	CI = 1210 - 1000 = 210/-	210/- - 200/- = 10/-	1210 - 1200 = 10/-
After 3 years	SI = [10%+10%+10%] of P = 30% of 1000 = 300 Note ①	= 1000 + 300 = 1300	$A_2 = 1000\,(1.1)^3$ = 1.331×1000 = 1331/- Note ②	CI = 1331 - 1000 = 331.	331 - 300 = 31/-	1331/- - 1300/- = 31/-

Points to be noted from the table:-

Note:- ①:- In SI, interest is calculated every year on principle only. constantly and added through out 'n' no. of years. constantly. [Its concept of SI].

↳ Every year interest is constant on principle if 'R' value is constant.

Note②:- Interest on interest is to be added extra and constant factor is multiplied through out 'n' no of years. [∴ 1.1 in the example table] [Its concept of CI]

Alternative [According to definition]:-

After 1 year:- $A_{2-1} = P + CI = 1000 + 10\%$ of 1000

$= 1000 + 100 = 1100$

After 2 yrs:- $A_{2-2} = \underset{P}{[1000 + 100]} + \underset{CI}{[10\% \text{ of } 1100]}$

$= 1100 + 110 = 1210$

[100 + 10% of 1100 = 2 yrs interest] = 210

[10% of 1100 is 2nd year interest] = 110

After 3 yrs:-:- $A_{2-3} = [1100 + 110] + 10\%$ of $[1100 + 110]$

$= 1210 + 121 = 1331$/-

[3 yr's Interest = 210 + 10% of 1210 = 331]

[3rd yr Interest = 10% of 1210 = 121]

After 4 yrs:- $A_{2-4} = [\underset{P}{1210} + 121] + 10\%$ of 1331.

$= 1331 + 133.1 = 1461.1$/-

[4 yrs Interest = 331 + 10% of 1331 = 461.1/-]

[4th yr Interest = 10% of 1331 = 133.1]

Note:③:- After 1 yr SI = CI [SI & CI are same] when P & R are constant and calculated annually.

Note④:- After any 'n' no. of years 'diff b/w CI & SI' is same as 'diff b/w A_2 & A_1' when P & R are constant & calculated annually.

[$A_2 \rightarrow$ Amount to be repaid on basis of CI
$A_1 \rightarrow$ " " " " " " SI.]

↳ Example Questions:-

Ex:-1:- $P = 25{,}000$, $R = 10\%$, $T = 3$ yrs $\Rightarrow SI = ?, A_1 = ?$

Ans:- $SI = \frac{25000 \times 3 \times 10}{100} = 7500$

(or)

$SI = (10\% + 10\% + 10\%)$ of $25000 = 30\%$ of $25k = 7500/-$

$A_1 = P + SI = 25k + 7.5k = 32.5k$.

Ex:-2:- A sum of money becomes doubled in how many years when R = 10% p.a in SI?

Ans:- $A_1 = P + SI = 2P$ [Sum doubled]

$\Rightarrow SI = P$

$\Rightarrow SI = \frac{\cancel{P} \times T \times 10}{100} = \cancel{P} \Rightarrow T = \frac{100}{10} = 10$ yrs.

(or) Assume P = 100%

$P \longrightarrow 2P \Rightarrow P\,100\% \overset{SI}{\xrightarrow{+100\%}} 200\%$

10% Interest → 1 yr

100% " " → $\boxed{10 \text{ yrs}}$